

Based on a detailed review of the provided data on mathematics professors at IIT Bombay, comprehensive, deeply refined research topics currently and recently pursued by the faculty include:

- Analytic number theory focusing on subconvexity bounds for L-functions, finite fields, automorphic forms, equidistribution problems, and representation theory of automorphic forms.
- Number theory research specializing in Hecke characters, Galois representations, mod p representations of p -adic groups, $SL(n)$ L-packets, toric periods, base change phenomena and conductors of local L-functions.
- Combinatorics and graph theory studying cascaded group testing, bounded fractional intersecting families, low-rank matrices from tournaments, symmetric designs, list coloring of triangulations, and sensor placement complexity for optimal observability.
- Hyperbolic conservation laws, computational fluid dynamics, numerical methods for Hamilton–Jacobi equations, nonlinear acoustics and shock wave reflection theory.
- High-dimensional statistics including change point detection, stochastic networks, time series analysis, free probability and random matrix theory.
- Algebraic geometry covering algebraic stacks, étale cohomology, intersection theory, Higgs bundles, geometric Langlands program, and moduli spaces.
- Fourier analysis and harmonic analysis on curves, extremizers for Fourier restriction theorems, and convolution operators on moment curves.
- Noncommutative probability and operator algebras focusing on multivariable operator theory, quantum probability, and operator spaces.
- Partial differential equations with emphasis on asymptotic theory, homogenization of differential equations, nonlinear hypoelliptic equations on manifolds, and eigenvalue problems for fourth-order operators.
- Stochastic control and mathematical finance investigating risk-sensitive control of diffusion processes, portfolio optimization under stochastic factors, and ergodic control of jump processes.
- Several complex variables, operator theory, spectral sets, dilation theory, and rational dilation failures on complex domains.

- Geometric analysis analyzing Ricci flow, curvature properties, spectral gap estimates, eigenfunction nodal sets, and heat diffusion on Riemannian manifolds.
- Design of experiments, generalized linear models, epidemiological modeling, optimal crossover designs, disease surveillance, and statistical inference for multivariate time series.
- Inverse problems and integral geometry including ray transforms, Schrödinger equations with time-dependent coefficients, range characterizations, and unique continuation principles.
- Algebraic topology and moduli spaces focusing on algebraic cycles, fundamental group schemes, and smash nilpotent cycles on various varieties.
- Arithmetic groups, Lie groups, thin groups, and representation theory with applications to number theory.
- Homogenization theory for wave propagation, reactive flows in porous media, and control theory applications related to hypocoercivity.
- Quantum ergodicity, spectral theory, Laplace eigenfunctions behavior under perturbations, uncertainty principles on manifolds, and applications to deep learning.
- Null controllability and stabilization of Navier-Stokes systems, viscoelastic flow control, compressible fluid models, evolution equations, and parabolic PDEs stabilization techniques.
- Partial differential equations related to tumor growth models, biharmonic plates, and numerical methods for non-selfadjoint indefinite systems.
- Complex variables, operator theory, rational dilation domains, and multivariate operator models associated with complex geometry.
- Combinatorial matrix theory, spectral analysis of random matrices, and linear algebra applications to graph theory and structured matrices.
- Mathematical finance and stochastic differential games focusing on portfolio optimization under partial information and risk-sensitive criteria.

This exhaustive collection captures the full depth and breadth of refined, nuanced mathematical research underway at IIT Bombay by the mathematics faculty, spanning pure, applied, computational, and interdisciplinary directions.

1. <https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/126871353/3370e023-aa27-4af3-afcf-4b2c3f0544f3/Untitled-document-9.pdf>